

Changes in expression and distribution of claudin 2, 5 and 8 lead to discontinuous tight junctions and barrier dysfunction in active Crohn's disease



Background Epithelial barrier function is impaired in Crohn's disease. Aim To define the underlying cellular mechanisms with special attention to tight junctions. Methods Biopsy specimens from the sigmoid colon of patients with mild to moderately active or inactive Crohn's disease were studied in Ussing chambers, and barrier function was determined by impedance analysis and conductance scanning. Tight junction structure was analysed by freeze fracture electron microscopy, and tight junction proteins were investigated immunohistochemically by confocal laser scanning microscopy and quantified in immunoblots. Epithelial apoptosis was analysed in terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate nick-end labelling and 4',6-diamidino-2-phenylindole staining. Results Patients with active Crohn's disease showed an impaired intestinal barrier function as indicated by a distinct reduction in epithelial resistance. As distribution of conductivity was even, focal epithelial lesions (eg, microerosions) did not contribute to barrier dysfunction. Instead, freeze fracture electron microscopy analysis showed reduced and discontinuous tight junction strands. Occludin and the sealing tight junction proteins claudin 5 and claudin 8 were downregulated and redistributed off the tight junction, whereas the pore-forming tight junctions protein claudin 2 was strongly upregulated, which constitute the molecular basis of tight junction changes. Other claudins were unchanged (claudins 1, 4 and 7) or not detectable in sigmoid colon (claudins 11, 12, 14, 15 and 16). Claudin 2 upregulation was less pronounced in active Crohn's disease compared with active ulcerative colitis and was inducible by tumour necrosis factor α . As a second source of impaired barrier function, epithelial apoptosis was distinctly increased in active Crohn's disease (mean (SD) 5.2 (0.5)% v 1.9 (0.2)% in control). By contrast, barrier function, tight junction proteins and apoptosis were unaffected in Crohn's disease in remission. Conclusion Upregulation of pore-forming claudin 2 and downregulation and redistribution of sealing claudins 5 and 8 lead to altered tight junction structure and pronounced barrier dysfunction already in mild to moderately active Crohn's disease.